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| **Fiscal Year:** | FY 2017 | **Task Last Updated:** | FY 11/28/2017 |
| **PI Name:** | Tash, Joseph S Ph.D. | | |
| **Project Title:** | Space Flight-Altered Motility Activation and Fertility-Dependent Responses in Sperm from Sea Urchin and Rodents | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Animal Biology: Vertebrate<br>(3) Animal Biology: Invertebrate | | |
| **Space Biology Cross-Element Discipline:** | (1) Reproductive Biology | | |
| **Space Biology Special Category:** | (1) Cell Culture | | |
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| **City:** | Kansas City | **State:** | KS |
| **Zip Code:** | 66160-0001 | **Congressional District:** | 3 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2012 Space Biology NNH12ZTT001N |
| **Start Date:** | 08/01/2013 | **End Date:** | 01/24/2018 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 1 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | Sato, Kevin | **Contact Phone:** | 650-604-1104 |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | ISS<br>NOTE: End date changed to 1/24/2018 due to change in PI (Ed., 1/31/18)<br>NOTE: End date changed to 11/30/2018 per F. Hernandez/ARC (Ed., 12/4/17)<br>NOTE: End date changed to 11/30/2017 per F. Hernandez/ARC (Ed., 10/24/16)<br>NOTE: End date changed to 11/30/2016 per A. Chu/ARC (Ed., 12/10/15)<br>NOTE: End date changed to 11/20/2015, per NSSC information (Ed., 8/10/15) | | |
| **Key Personnel Changes/Previous PI:** | UPDATE NOTE (June 2018): Project continues with new PI Gustavo Blanco --"Space Flight-Altered Motility Activation and Fertility-Dependent Responses in Sperm from Sea Urchin and Rodents (PI: Blanco)," with grant number 80NSSC18K0522. Originally the grant had continued with same grant number as noted in January 2018 below note.<br>NOTE (January 2018): Project continues as "Space Flight-Altered Motility Activation and Fertility-Dependent Responses in Sperm from Sea Urchin and Rodents (PI: Blanco)" with the same grant number NNX13AM62G due to retirement of Principal Investigator Joseph Tash, Ph.D. | | |
| **COI Name (Institution):** | | | |

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| **Grant/Contract No.:** | NNX13AM62G |
| **Performance Goal No.:** | |
| **Performance Goal Text:** | |
| **Task Description:** | NOTE from June 2016 report: Nearly one year ago, we amended the grant proposal to include cryopreserved bovine and human semen, while eliminating the use of sea urchin and rat species.<br>Sperm motility and acquisition of the capacity to fertilize are the most fundamental components of male reproduction necessary for species survival on Earth or in space. Previous space shuttle flight experiments that we performed demonstrated that the signal transduction (i.e., m protein phosphorylation) required for activation of sea urchin sperm occur more rapidly in microgravity than at 1G. In contrast, we found that signaling in sperm triggered by egg chemotactic factors is delayed in microgravity compared to 1G. No flight-capable microscope was available during those experiments to determine whether the signaling changes were in fact coupled to differences in motility under the microgravity versus 1G conditions. Our hypothesis is 1) that sperm motility (sea urchin and mammalian) is activated more rapidly in microgravity than on Earth, and 2) that the response of sea urchin and mammalian sperm (motility hyperactivation, and morphological changes) to fertilization activating conditions, as well as the underlying metabolic and regulatory mechanisms, are delayed or blocked in microgravity. In this regard, if the corresponding activation and changes in sperm motility do not occur, then fertilization cannot occur and reproduction will fail. Thus, it is essential to assess whether the critical component of sperm function, motility, and the underlying signalling and metabolic support are altered in space flight conditions. We propose to use existing flight hardware to activate sea urchin and rodent sperm in microgravity, and quantify multiple parameters of motility and morphology, combined with detailed metabolomic pathway analysis. Previously utilized methods for preparing inactive sea urchin sperm for subsequent activation in space flight will be adapted for mouse and rat sperm. We will also test whether activating peptides from sea urchin eggs, or mammalian capacitating conditions trigger temporal or magnitude changes in sperm motility, morphology, as well as essential regulatory glycolytic and energy metabolic pathway markers (using novel metabolomic analysis) in microgravity versus 1G. Most importantly, this comprehensive project will lead to the first direct in vitro assessment of canonical fertility-critical functional processes of mammalian sperm in microgravity that are identical to many human in vitro clinical and research laboratory assessments. These experiments are 1) part of the Rapid Turn-around Space Flight Experiments Research Emphasis, and 2) address the National Research Council (NRC) 2011 Decadal Survey to discover and characterize underlying mechanisms of adaptation to the space flight environment, and to determine cellular and organismal mechanisms that regulate and sustain reproduction and development in space. In this regard, the results of these experiments will provide important fundamental reproductive biological data to support NASA's goal to conduct multigenerational studies of mammals in space. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Previous studies indicate that reproductive functions in sperm that occur on the ground also occur in space flight. However, the timing of signal transduction changes upon initiation of sperm motility is more rapid, yet, the fertility-related responses of sperm to the presence of egg chemotactic factors are delayed. This suggests that sperm fertility may be affected resulting in subfertility or infertility in males during and post flight. There is insufficient data available from humans to assess this possibility due to low n's and HIPPA concerns.<br>The flight logistics for the new SpaceX vehicle require a wider window of time between sample handover and subsequent experiment activation by crew on the International Space Station (ISS). New procedures for extending the viability of quiescent sperm are being developed for both sea urchins and rodents (rats). |
| **Task Progress:** | We have successfully finalized the experimental design of the flight project and the hardware to be used both in flight and corresponding ground controls. In addition, we have successfully collected both human and bovine pooled samples in the space flight compatible hardware. We have also done extensive thaw time testing for these samples as well as the media in which they will be incubated in order to determine pre-incubation procedures. We have obtained toxicology data necessary for flight manifest. Working extensively with BioServe, we have assembled procedures for flight incubation and addressed issues that have arisen in this regard.<br>We conducted a highly successful EVT (experiment verification test) Oct 18-23 at BioServe’s facilities at the University of Colorado-Boulder. A formal report of the EVT was prepared by Bioserve, approved by me, and submitted to the NASA project scientist Fathi Karouia. We met with Fathi at the American Society for Gravitational and Space Research (ASGSR) meeting in October with Bioserve and my lab team in attendance, and discussed the EVT overall, the EVT report, and made plans to move forward towards the science requirements review (SRR). We are now analyzing the EVT materials with a focus on assessing the EVT results with respect to the success criteria table in preparation for the SRR. The initial results of the EVT were presented in a oral platform session at ASGSR. Running a full EVT allowed us to refine the procedures as we went along and generate high quality samples that will likely show that the success criteria will be met with a high level of quality.<br><br>UPDATE NOTE (June 2018): Project continues with new PI Gustavo Blanco --"Space Flight-Altered Motility Activation and Fertility-Dependent Responses in Sperm from Sea Urchin and Rodents (PI: Blanco)," with grant number 80NSSC18K0522. Originally the grant had continued with same grant number as noted in January 2018 below note. See project with PI Blanco for future reporting.<br><br>NOTE (January 2018): Project continues as "Space Flight-Altered Motility Activation and Fertility-Dependent Responses in Sperm from Sea Urchin and Rodents (PI: Blanco)" with the same grant number NNX13AM62G due to retirement of Principal Investigator Joseph Tash, Ph.D. |
| **Bibliography Type:** | Description: (Last Updated: 01/24/2022) |

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| **Abstracts for Journals and Proceedings** | Yarns EM, Stodieck LS, Zea L, Tash JS. "Flight Implementation of In Vitro Capacitation Studies of Cryopreserved Human and Bovine Sperm on the ISS." Presented at the 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017.<br>33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017 |